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Conductance/impedance techniques for microbial assay

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17.1 Introduction

Over a century ago Stewart (1899) 'found that the electrical conductance of a medium in which bacteria were growing increased significantly'. He showed that the change in conductance was proportional to the concentration of the electrolytes produced, and deduced that the conductance change was thus proportional to the concentration of active bacteria and so could be used as a measure of microbial numbers (counts). He speculated that 'sufficiently great and constant differences might be observed between different kinds of bacteria when grown in the same or different liquid media to enable us to use the method as a supplement to our present means of diagnosing between nearly related forms', in other words, as a means for identifying bacteria.

It is interesting to note the design of the equipment he used and the experimental conditions he thought desirable. He measured the electrical resistance of a cell containing the medium by means of a Wheatstone bridge under alternating current conditions using a 'telephone' for zero detection. It was necessary to prevent evaporation of the medium and to maintain constant temperature during growth so as to avoid artefactual changes in resistance. Stewart though it desirable to use 'tubes furnished with platinum electrodes fused through the glass, or perhaps passing through the plug' to avoid contamination.

He was aware of the essential features that have made modern impedance methods so successful today but could not develop the methodology because of the limitations imposed by the available technology of his time. Despite sporadic interest (see Easter and Gibson 1989) the subject lay virtually dormant until Ur and Brown (1975) and Cady (1975) drew attention to the potential application of electrical impedance measurement and designed apparatus for displaying its change during bacterial growth. They demonstrated the distinctive effects of different organisms grown in a given broth and the influence of different growth media on the shape of the curves of impedance or conductance with time since inoculation. The possibilities for the rapid automation of microbiological tests were clearly indicated by these authors.

In this chapter, the current rapid microbiological methods for detection and identification of micro-organisms are described, an outline of the underlying principles

of impedance measurement for cultures is given along with the factors influencing sensitivity and a survey of the growth analysers available commercially. The use and usefulness of the method in food quality control is discussed.

Automated conductance/impedance methods have become well established in analytical microbiology and feature in reviews (Silley and Forsythe, 1996) and working manuals as accepted procedures especially for screening large numbers of samples with minimal technical effort yielding results within a day (Boer and Beumer, 1999). Samples are inoculated into a suitable medium in an electrode-cell that is connected to a measuring device. When a pre-selected change in conductance/impedance and/or its rate of change has occurred relative to a control, microbes have been detected. The time since inoculation can be calibrated with the number of microbes inoculated for 'counts' or taken as indicative of the presence of particular microbes. The sensitivity, that is, the lowest number of microbes detected, is high; only one cell needs to be present initially as it multiplies in the growth medium until the detection threshold of the instrument is attained. The specificity of an assay depends on the constraining influences of the medium, as in traditional microbiology. The rate of change of conductance/impedance can be related to the microbial growth rate.

Impedance techniques have found successful practical applications in the food industry, and commercial instruments are available to the industrial quality control laboratory. They always feature in reviews and symposia on rapid methods and are reported to be high on microbiologists' 'wish list' because of their ease of use and speed of result. In practical terms, the assay consists of a simple inoculation of a microbiological medium in a proprietary cell, and attachment to an instrument which takes regular readings and presents results on up to hundreds of simultaneous assays on a computer screen in about a quarter of the time taken by conventional tests. No other rapid system is so simple nor has such a high sample capacity.

17.2 Rapid microbiological methods: an overview

There are two main objectives in food microbiology; firstly to determine the total load or numbers of microbes in a sample, and secondly to determine the presence or absence of a particular microbial species, usually pathogens or related types used as their indicators. The total load is determined as the total viable count (TVC) or the number of colony forming units (cfu) or particles (cfp), and relates to quality and stability of the food. Enumeration of specific groups is associated with safety or assessing the risk of food poisoning. The results are not always absolute, being influenced by the method and incubation conditions used. There are also assays for specific toxins or poisons produced by microbes but these are not considered here as they are not measured in growth assays. The growth medium after conductance assay can be used as the primary testing material for such and any other assays.

The time taken to complete conventional microbiological assays has changed little as it depends on the time taken for microbes to grow and multiply so as to give a visible colony in the surface of a nutrient solid medium, colour change by the metabolism of specific chemicals, or obvious turbidity in a liquid medium. Much effort has accordingly been devoted to shortening assay times and to replacing the visible end points with alternative measurements. The procedures used in the conventional assay involve the following stages. First a sample, normally 10 to 25 g, is homogenised in a buffer or medium to release the microbes from any solid material and to dilute out any natural anti-

microbial compounds. Then the extract is further diluted in serial decades in buffer and inoculated in small volumes, 10–1000 μl either onto the surface of a medium, or into a warm liquid medium that solidifies on cooling, or into a broth. After incubation for, say, 1–7 days, visible colonies that have grown on a solid medium are counted on the most suitably diluted sample (yielding 30–300 colonies) or the turbidity of a liquid medium are noted. Further tests are often done especially to confirm the identity of pathogens or organisms indicative of their presence. Thus much of the routine of food microbiology is laborious, technically demanding and slow to yield results. Some of the steps have been mechanised, for example using diluter/spreaders and colony counters, but the overall assay times have altered little, even with the advent of improved and more selective media.

The birth of rapid methods in microbiology dates from 1973 when the first symposium on 'Rapid Methods in Microbiology and Immunology' was held (Héden and Illeni 1975a; 1975b). Many new approaches were described, some of which became available commercially, while others were discarded. Apart from mechanisation of various manual steps in assays, the main techniques for which there are instruments available are those based on electrical methods (conductance or impedance measurement, electrochemical assay of the culture broth), on chemical methods such as the determination of bacterial adenosine triphosphate (ATP), on colorimetric measurement of dye reduction, on filter techniques (direct epifluorescent filter (DEFT), hydrophobic grid membrane), cytometry, and on agglutination methods (immunological, limulus amoebocyte lysate (LAL)). Recently, new techniques have appeared based on nucleic acid technologies. These and others have been reviewed in Adams and Hope (1989), Huis in't Veld *et al.* (1988), Stannard *et al.* (1990), Bolton and Gibson (1994) and Betts (2000).

Of all the methods, the electrical assays have been most widely accepted by the food industry. As described later, the net metabolic activity of the culture is represented by conductance assays and this correlates well with the microbial load in the material. Impedance includes a capacitance component which can be influenced by microbial and other biomass, whether living or dead. Specific media can be used for particular organisms. The higher the microbial load, that is high numbers, the shorter is the assay time. Results can be obtained in 2–4 h. Conversely, foods with low numbers such as those that have undergone a heat treatment require longer incubation times. For the detection and counting of specific pathogenic bacteria, substrates used uniquely by the target bacteria of inhibitors to which all but the target bacteria are sensitive can be included in the medium to increase the response by the pathogen and minimise the growth and contribution by other microbes. The instruments are simple to use; an extract is made, added to the measurement cell and connected to the instrument, which then monitors the assay and reports the result. Up to 500 samples can be examined simultaneously on current machines. Thus extensive sampling can be done with few staff, so justifying the capital outlay. Indeed, the payback time for equipment has been less than a year in many instances. The output data are accepted by many retail food companies in place of conventional data. The instruments have a particularly useful role in product and process development and risk analysis.

All living cells contain ATP (adenosine triphosphate), the high energy intermediate that powers most energy-consuming reactions. The test depends on the reaction between ATP and the enzyme luciferase, producing light which is measured photometrically with a claimed sensitivity down to 10^{-16} mol 1^{-1}. It is a rapid test, taking < 1 hour to complete. The problem with most foods is that they, like the microbes present, contain ATP, and that the ATP content of microbial cells is variable depending on their nature,

bacterium, yeast, spore, etc. and their physiological state and the assay has no specificity. Manufacturers have produced reagents to selectively destroy somatic (food) cell ATP, by using detergents to lyse the cells and ATPase to destroy the ATP. However, there is increasing use of ATP measurement as a hygiene test for the cleanliness of food production areas. For this, the total ATP is measured as food residues contain far more ATP than the bacteria and a pass/fail standard can be set. Also, most foods contain more ADP (adenosine diphosphate) than ATP and ADP can be converted to ATP enzymatically. Both these developments increase the sensitivity of the assay. The assays are useful in materials which have little intrinsic ATP such as cooked meat products, carbonated beverages and fruit juices. The proceedings of a symposium on the subject have been published (Stanley *et al.* 1990). Despite the widespread use of ATP technology, there have been no international trials, by bodies such as AOAC, to assess the method or to compare kits from different manufacturers.

Automated microscopy methods have been commercially produced. The main problem relates to the small size of microbes, their relatively low density even in spoiled foods such that the microbial flora represent < 0.01% of the material. Accordingly, the microbes must be treated to stand out from the background mass. Dyes – coloured and fluorescent – are used. One example is the DEFT (Direct Epi-Fluorescent Test), originally devised for the liquid milk industry (Pettipher 1983) to provide a rapid test to back up raw milk payment schemes. It depends on the uptake by cells of acridine orange. Viable cells fluoresce orange under ultraviolet light owing to their ribonucleic acid (RNA) content, whereas dead or non-growing cells fluoresce green owing to interactions of the dye with deoxyribonucleic acid (DNA). Samples are concentrated by filtration, stained and viewed microscopically. Image analysis equipment can be used to automate counting. The Bactoscan is a fully automated variant. These tests have been applied mainly in the dairy industry on products which have not undergone a heat treatment which could give misleading results, and other foods which can be easily filtered. The sensitivity is only ~10^4 /ml. Procedures have been devised for preparing suitable filtrates from solid foods, but correlations with conventional assays have often been too low. The hydrophobic grid membrane technique also relies on filtration of a sample or an extract such that the microbes are deposited on a special filter membrane on which a grid of hydrophobic lines is imposed. The filters are incubated until each microbe has multiplied to occupy a square, facilitating simple automated counting techniques. There are procedures for handling various foods, incorporating filter aids, using enzymes to destroy spurious particles, etc. Selective media can be used and the grids stained with dyes or immunochemicals after incubation. Flow cytometry techniques have been reported. The sample containing microbes is injected into a stream of fluid which then passes a sensor where each particle is detected. Protocols are being developed and good results are reported for some foods although the sensitivity is poor.

Immuno-chemicals have been investigated for use in food microbiology. Microbes are antigenic and can stimulate the production of antibodies when injected into animals. Polyclonal antibodies, which react with a broad range of antigens, and monoclonal antibodies which are highly specific to particular antigenic structures have been used in reagent kits for the detection and identification of specific types of bacteria, their surface structures and toxins. The antibodies are tagged to assist in measurement of the antigen-antibody complexes. The most sensitive labels are radioactive isotopes but these cannot be used in food production environments. Fluorescent antibodies labelled in particular with fluorescein or umbelliferones are most common. Experience has shown that such reagents have a role in identification of isolates, especially after culture as the assays are

insensitive to low numbers of antigens. There have been reports of the incorporation of antibodies, produced in plants, in packaging films. They show visible changes on reaction with the target microbes. Immuno-magnetic reagents have been developed for capturing and concentrating pathogenic bacteria and have become essential steps in protocols for *E. coli* O157 detection. They permit larger volumes of extract and hence of food to be screened. Ogden *et al*. (2000) show a 9-fold increase in sensitivity by such means.

Nucleic acid hybridisation technologies have been developed for food use. They have a serious drawback in that they do not usually distinguish between living and dead cells when applied in their standard format. Target DNA is amplified using the polymerase chain reaction (PCR) and then short nucleotide probes, specific for the organism sought, are hybridised. The actual assay takes a few hours but usually cultures have to be isolated from the food, grown and then used so that the total assay time is 40–50 hours, no more rapid than older technologies, but highly specific. Some manufacturers have promoted the use of combined technologies – screening out negative cultures by conductance and confirmation of positives by the use of PCR kits.

17.2.1 Impedance as an indicator of microbial load

With impedance techniques, an increase in the conductance, capacitance or admittance (having both conductance and capacitance components) can be measured as microbial growth proceeds in the medium. The conductance increase is due to charged metabolites being produced by the microbes from uncharged substrates, and these will contribute to the capacitance increase. An increase in the measured capacitance can also be observed when biomass attaches to the electrodes. Such an increase in capacitance would occur later in the growth cycle and could be due to cells already inactivated or dead and perhaps due to tissue cells from the foodstuff. The conductance increase, on the other hand, always indicates the presence of active microbial cells and is observed earlier in the growth cycle. Change of pH of the medium caused by metabolism without growth can change the conductance and capacitance.

Conductance monitoring is therefore the method preferred by the author for most microbial assays. Capacitance measurement is often used for the assay of yeasts and moulds which do not produce ionic metabolites from standard growth media. With such capacitance methods, care must be taken to avoid errors due to artefacts related to the condition of the electrodes. It should be noted that the frequency of measurement will affect the choice between conductance or capacitance methods. This chapter is based largely on experience gained in the development of and studies with the Malthus Analyser, which operates at a frequency of 10 kHz. Other commercial instruments operate at frequencies between 2 and 10 kHz. The Bactometer, for example, operates at 2 kHz, and for this frequency the capacitance change with microbial numbers is more pronounced.

The choice of capacitance or impedance measurement instead of conductance monitoring will be discussed in more detail in Section 17.4. The description of impedance techniques in this chapter will refer to conductance measurement unless noted otherwise. However, in order to determine the conductance of the growth medium accurately, it is necessary to examine the impedance of the system consisting of the fluid and the electrodes. In certain cases, and with appropriate instrument design, it is also possible to measure admittance as an approximation to conductance.

17.3 Principles of electrical conductance methods

17.3.1 Impedance and its component variables

Consider the current that flows between two electrodes immersed in a fluid such as microbiological media when an alternating potential is applied, there are three components that limit its flow. These are: (1) the true resistance R_s of the fluid; (2) a capacitance C_{ox} in series with a resistance R_{ox} resulting from the presence of an oxide layer at the surface of each electrode; and (3) a capacitance C_D in series with a resistance R_d generated close to each electrode as a consequence of the alignment of polar dipoles in the fluid in a double charge layer (Fig. 17.1(a)). For convenience the electrode capacitances are lumped together as a single series capacitance C_{se}. The resistance values R_{ox} and R_D are generally small in relation to R_s and may be neglected in some less critical applications'. In more demanding circumstances, such as the investigation of early growth characteristics, the need for the absence of an oxide layer and minimum electrode polarisation indicates the use of platinum electrodes. The combined effect of components (1), (2) and (3) as 'seen' by a measuring system is therefore an impedance consisting of a single capacitance C_{se} in series with a single resistance R_s (Fig. 17.1).

The instantaneous current i that flows through a pure resistance R_s is proportional to, and in phase with, the applied sinusoidal voltage V such that

$$i = V/R_s$$

$$= (|V|) \cos 2\pi ft/R_s \quad (1)$$

where $|V|$ is the amplitude (peak value) of the voltage, f is its frequency and t is time.

The mean current $|i_m|$ averaged over a complete cycle is

$$|i_m| = |V_m|/R_s \quad (2)$$

where $|V_m|$ is the mean voltage modulus averaged over a complete cycle.

For a pure capacitance C_{se} it is the charging current that determines i, and because the maximum current flows when $V = 0$ and no current flows when $V = |V|$, i flows in advance of V by an angle $\pi/2$. Hence

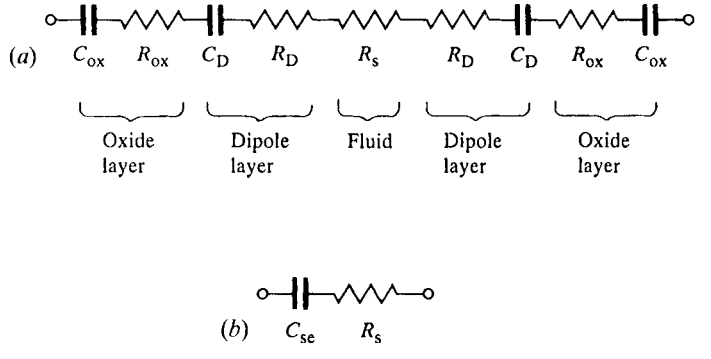


Fig. 17.1 Impedance measured across two metal electrodes immersed in a fluid: (a) impedance components for the fluid (R_s), the metal oxide (C_{ox}, R_{ox}) and the dipole double charge layer (C_D, R_d) (*b*) equivalent circuit for R_{ox}, $R_D \ll R_s$.

$$i = C_{se}(dV/dt)$$

$$= C_{se}\frac{d}{dt}(|V| \cos 2\pi ft)$$

$$= 2\pi f C_{se}|V| \cos (2\pi ft + \pi/2)$$

Therefore

$$i_m = 2\pi f C_{se}|V_m| \quad (3)$$

$$= |V_m|/X_{se}$$

where $X_{se} = (2\pi f C_{se})^{-1}$ is termed the *reactance* of the capacitance C_{se} and has the dimensions of resistance.

The effect of placing R_s and C_{se} in series is to reduce the angle by which the current leads the voltage measured across the terminals to a value such that $0 < \phi < \pi/2$. ϕ is termed the 'phase angle'. The total impedance is a vector quantity Z which is the resultant of R_s and $-X_{se}$. Note that the negative value of X_{se} indicates that the reactance vector leads the resistance vector by an angle $\pi/2$ and that it has no component on R_s.

Referring to Fig. 17.2, we see that

$$R_s = |Z| \cos \phi \quad (4)$$

$$-X_{se} = |Z| \sin \phi \quad (5)$$

so that

$$Z = (R_s^2 + X_{se}^2)^{1/2}$$

$$= [R_s^2 + (2\pi f C_{se})^{-2}]^{1/2} \quad (6)$$

17.3.2 Cell design and geometry and composition of electrodes

The shape of the vessel and the size and disposition of the electrodes are not critical in the design of impedance cells unless high resolution is required. The materials of which they are

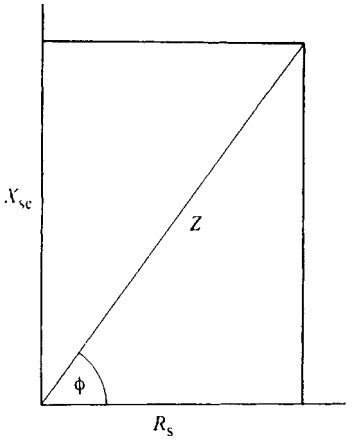


Fig. 17.2 Vector representation of impedance Z for R_s and C_{se} placed in series: $X_{se} = (2\pi f C_{se})^{-1}$ and ϕ is the phase angle.

constructed should not contaminate the growth medium, should not affect the growth of microbes, should withstand sterilisation and, for reusable cells, should be easily cleaned. Electrodes a few square millimetres in area, separated by a distance of 3 to 50 millimetres, will result in easily measurable impedances for common culture media. In general, it is desirable to position both electrodes in a vertical plane above the base of the cell to prevent detritus from coating them during the period of measurement. Alternatively the electrodes may be coated in agar to eliminate direct deposition, or the viscosity of the medium may be increased with dextran (Curtis *et al.* 1985). Nevertheless, these requirements may be relaxed if the time of detection of some high level threshold of growth, say 5×10^6 cfu ml^{-1}, is the desired measurement; then electrodes protruding through the base or situated in the plane of the base form an acceptable arrangement. It is essential that electrical connections to the electrodes consist of low resistance self-cleaning contacts since variations of more than $0.01\,\Omega$ may be easily detected.

The electrode material must of course be non-toxic to microbes and must be corrosion-resistant. A cheap option is stainless steel but, for the best calibration between conductance change and change in the numbers of bacteria is necessary, the requirements are more demanding. To obtain such linearity it is preferable to measure R_s directly. Attention must be paid to minimising electrode polarisation and to avoiding oxide formation on the electrodes, since these produce variable and unpredictable impedance effects. Schwann (1963) has investigated the measurement of biological impedances in considerable depth, and has concluded that a platinum surface coated with platinum black offers the lowest possible interfacial impedance but that a plain platinum surface is acceptable for the measurement of all but the lowest impedances. All other materials are inadequate in relation to platinum and their deficiencies become more pronounced as the measurement frequency is reduced. For example, Gibson and Jason (1993) showed that the impedance Z was no longer similar to R_s at low values with highly polarised electrodes and at frequencies below 10 kHz unpredictable fluctuations in C_{se} occurred.

It is evident that at 10 kHz the capacitance C_{se} varies over a small range only (from $0.87\,\mu$F to $2.41\,\mu$F for platinum electrodes), while the true resistance of the fluid RS varies from $22.5\,\Omega$ to $938.1\,\Omega$ under all conditions. Also the impedance Z approximates quite closely to R_{s} for large values of R_{s} but, in the case of highly polarised electrodes, diverges seriously for the lower values. The use of electrodes composed of metals other than platinum, or the reduction of measuring frequencies below 10 kHz, leads to errors in assuming Z and R_{s} are equivalent. Unpredictable fluctuations in C_{se} will also give rise to a certain lack of discrimination if impedance is measured instead of resistance. Thus, if the requirements are exacting, it is preferable to measure R_{s} when the medium is highly conducting. Otherwise, the time taken for a positive assay result is increased, perhaps by 10–20%.

17.3.3 Relationship between test cell conductance and bacterial growth

A plot of the change in conductance of a bacteriological medium against time since inoculation is shown in Fig. 17.3. There is an initial period, the lag phase, in which cells metabolise but the bacterial numbers remain unchanged while the bacteria adapt to the medium. As a rule, there appears to be no detectable change in the conductance of the growth medium during this phase. After cell division commences, more metabolites, mainly ionically charged, pass into solution and the conductance increases provided the sensitivity of the detection instrument is exceeded. The introduction of ions enhances the conductance G_{s} of the medium, and the enhancement δG_{s} is proportional to the growth in population density $n - n_o$ and the mobility of the ions. During each phase of growth, a

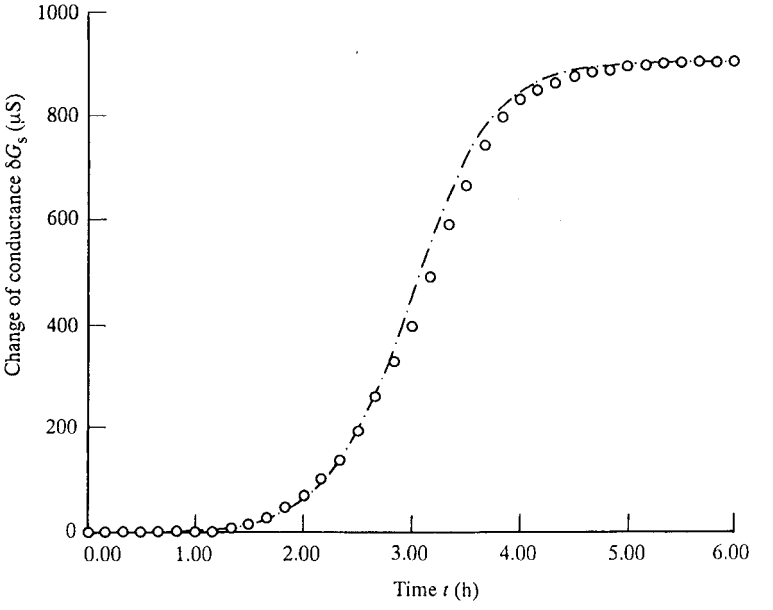


Fig. 17.3 Change of conductance δG_s with time after inoculation of nutrient broth (Oxoid CM1) with *E. coli* (8×10^5 cfu ml^{-1}). The broken line represents equation 16 in which KG_s/k is 1*pS* (cfu ml^{-1})$^{-1}$.

bacterial cell probably produces a fixed number of ions per generation. Thus δG_s may be regarded as a conductance shunting the initial conductance of the medium G_0, as illustrated in Fig. 17.4, its value being

$$\delta G_s = 1/R_s - 1/R_0 \tag{7}$$

where R_0 is the initial value of the resistance R_s of the medium. The unit in which G_s is measured is the siemens (S). An example showing the linearity of the relationship to bacterial numbers is presented in Fig. 17.5.

Consider a cell containing a medium of initial specific conductivity k, in which electrodes of area A and effective separation l are immersed, then

$$G_s = kA/l \tag{8}$$

The production of a single bacterial cell in unit volume of medium is associated with an increase of specific conductivity K, so that

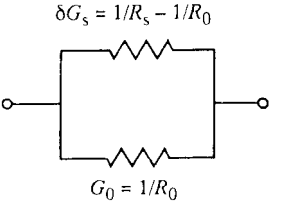


Fig. 17.4 Shunting effect of conductance increase due to bacterial growth on initial broth conductance.

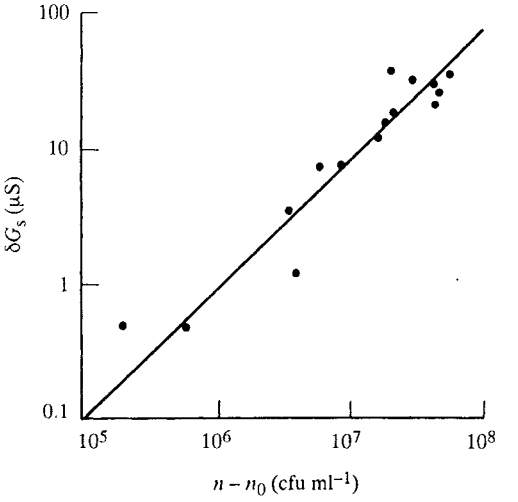


Fig. 17.5 Change of conductance δG_s with increase in cell population density $n - n_0$ for *E. coli* in CM1 broth at 37°C.

$$\delta G_s = (KA/l)(n - n_0) \tag{9}$$
$$= (KG_s/k)(n - n_0)$$

For Fig. 17.5, the constant of proportionality KG_s/k is approximately 1 pS (cfu ml^{-1})$^{-1}$, so that the value of K may be obtained by substituting the value of k for the nutrient broth, Oxoid CM1, (88 S cm^{-1}) and G_s measured at l0 kHz (derived from Table 17.1, $R_s = 1/G_s = 279\,\Omega$):

$$K = 2.45 \times 10^{-8}\text{S cm}^{-1}(\text{cfu ml}^{-1})^{-1}$$

The magnitude of K generally lies within ±20 per cent of this value for a wide range of bacteria in this medium and is of the same order for bacteria in other media (Gibson and Hobbs 1987).

Such assays are referred to as direct conductance assays. There is also a technique known as indirect conductance in which the electrodes are kept separate from the growth medium (Owens *et al.* 1989). The electrodes are in an alkaline solution. As the microbes metabolise, they produce carbon dioxide (CO_2). Initially the gas is dissolved in the medium and then equilibrates into the headspace. It diffuses into the alkaline solution, usually NaOH or KOH, where it forms a carbonate ion. There is a difference in

Table 17.1 Influence of measuring frequency on series impedance (CM1 broth, pH 7.2, 37°C, 0.5 g NaCl per 100 g H_2O)

Frequency (kHz)	Z (Ω)	R_s (Ω)	X_s (Ω)
10	279.0	278.6	14.7
5	280.1	278.6	29.4
2	288.1	278.6	73.5
1	315.0	278.6	147.0
0.5	405.0	278.6	294.0

conductivity and conductance between the hydroxyl and carbonate ions that is measured. There is no capacitance component.

17.3.4 Bacterial number resolution of analysers

All analysers used in the food industry employ digital techniques to measure either impedance Z or its inverse, admittance Y, or, when true linearity may be required, either resistance R or conductance G. Their general design is described below.

The numerical resolution of an analyser is limited by the resolution ΔV of the analogue-to-digital (A/D) voltage converter employed to display whichever variable is being measured. If, say, R_s is the measured variable and R_{fs} is the full-scale range corresponding to the full-scale voltage range, then the smallest value of resistance that can be resolved is

$$\Delta R_s = R_{fs}/2^b$$

where the voltmeter has a range of b bits. The corresponding resolution of conductance is therefore

$$\begin{aligned}\Delta G_s &= -\Delta R_s/R_s^2 \\ &= -R_{fs}/R_s^2 2^b \end{aligned} \tag{10}$$

Substituting this value into equation (9) and denoting Δn as the limiting numerical resolution corresponding to ΔG_s no change, we have

$$\Delta n = kR_{fs}/KR_s 2^b \tag{11}$$

Equation (11) shows that the resolution is determined by the ratios k/K and R_s/R_{fs} and by b. The dominant factor controlling k/K is the concentration of the salts in the medium.

The resolutions for various salt concentrations, bit values and ratios R_s/R_{fs} are shown in Table 17.2. To achieve the highest possible resolution for direct assays, three conditions must be met:

- low specific conductivity of growth medium
- value of ratio R_s/R_{fs} close to unity
- high resolution of analogue-to-digital converter.

17.3.5 Temperature control

The temperature coefficient of conductance of media is very high. For an isotonic medium it is 0.016 K^{-1}, for example, so that the conductance of a cell of say 3000 μS would change by 0.48 μS for each change of 0.01 K. Clearly it is essential to maintain cell temperature constant to within a few milli-degrees of a desired value. This is readily achieved by immersing cells in a thermostat controlled water bath (Fig. 17.9(a)) or by placing them in deep holes bored in an aluminium or copper block similarly controlled. One manufacturer (Bactometer) employs an air incubator (Fig. 17.9(c)) and a system of balanced pairs of cells, one of which (as described below) is a sterile control to minimise temperature effects, but the practice is only satisfactory for low resolution determinations. Another manufacturer (BacTrac) uses 4-terminal measurements. Both the water bath and the aluminium block can control cell temperatures within ±1mK and fluctuations and drift in conductance measurement can be lower than 0.1 μS h^{-1}.

Table 17.2 Numerical resolution of conductance analyser

NaCl molarity	k/K[a] (cfu ml^{-1})	Bit no. b	Resolution (10^6 cfu ml^{-1}) for R_s/R_{fs} = 0.1	0.3	0.6	0.9
0.146M	3.59×10^9	14	2.19	0.73	0.37	0.24
		16	0.55	0.18	0.09	0.06
		18	0.14	0.05	0.02	0.01
		20	0.03	0.01	0.01	0.004
0.060M	1.53×10^9	14	0.94	0.31	0.16	0.10
		16	0.23	0.08	0.04	0.03
		18	0.06	0.02	0.01	0.006
		20	0.01	0.005	0.002	0.002
0.028M	7.18×10^8	14	0.44	0.15	0.07	0.01
		16	0.11	0.04	0.02	0.01
		18	0.03	0.01	0.005	0.003
		20	0.007	0.002	0.001	0.001

[a] Inverse of normalised conductivity increase.

17.4 Capacitance versus conductance measurement

17.4.1 Selection of display

There has been much controversy regarding which is the best electrical variable to be displayed. The key data are found in the papers by Richards *et al.* (1978) and Firstenberg-Eden and Zindulis (1984), and Kell (1987). Richards *et al.* (1978) measured series conductance G_s at 10 kHz. They stated: 'Experimentally it was found that C was subject to fluctuations which did not correlate well with any measured variable, such as temperature. These more or less random fractional changes in C were of the same order of magnitude as the systematic fractional changes in G which could be ascribed to bacterial growth; and it became clear that measuring C was most unlikely to be a reliable means of observing bacterial growth.' This work was confirmed using a commercial Malthus instrument. (See, however, Sections 17.4.2 and 17.2.1 on the changes later in the growth cycle and on the influence of frequency.) Careful examination of Fig. 17.6 given by Firstenberg-Eden and Zindulis (1984) show that conductance is the electrical component that changes earliest; they were using a test rig rather than an actual instrument.

There are no systematic changes in C_{se} when changes in G_s are already observed. This was evident for both bacteria and yeasts. This is not to imply that C_{se} does not change later but, for early detection of growth, C_{se} is a less sensitive indicator than G_s. Kell (1987) indicates that cells no longer viable contribute to the magnitude of the capacitance C_{se}, so that dead or nearly dead cells contribute to C_{se}.

In routine quality control the magnitude of the electrical signal should correlate with change in microbial number at all times. For example, food spoilage or deterioration leading to its rejection by consumers is based on the activities of microbes rather than their biomass, as many foods such as cheese or yoghurt contain large numbers of non-metabolising non-growing cells. There are no reports of false decisions based on changes in conductance, but Jarvis (1985) showed that wrong decisions were made when a change in capacitance obtained with fruit materials was found. The product was in fact sterile and the change was ascribed to the 'unauthorised use of an acidulant'. If added acids can give continuous drift of the measured variable due to corrosion of the electrodes, then a more

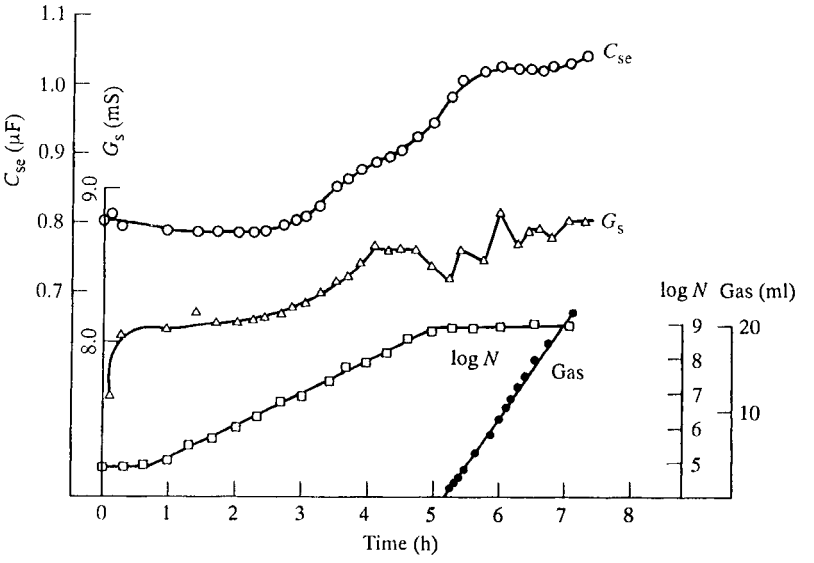


Fig. 17.6 The effect of the growth of *E. coli* in brain heart infusion on capacitance C^{se}, conductance G_s, number N of bacteria and gas evolution (from Firstenberg-Eden and Zindulis 1984, by courtesy of the authors and publisher).

robust display variable is desirable, or totally inert electrodes. Capacitance measurement has had an important role to play in the detection of some yeasts and moulds, but the development of indirect conductance methods and of media formulations enhancing conductance signals (such as described by Connolly *et al.* 1988; Adams *et al.* 1989) is generally a safer route forward; simultaneous measurement of C_{se} and G_s may be safer than G_{se} alone in certain applications.

17.5 Instrument design

17.5.1 Method of measurement

There are several ways in which the conductance/impedance of media may be measured. The most direct method employs a four-terminal bridge measurement (Kent and Jason 1975) in which the influence of interfacial electrode impedances are eliminated and the impedance of the medium is measured directly. At audio frequencies (20 Hz to 20 kHz) the impedance of culture media is entirely resistive, so that it is possible to devise an arrangement which enables each of the component variables to be measured directly as desired while at the same time permitting the operation of a simple multiplexing arrangement.

Essentially it is necessary to measure the resistance R_s, its equivalent conductance G_s or capacitance C_{se}. The resultant impedance Z or admittance Y ($Y = 1/Z$) is often measured as well as, or instead of, R_s or G_s. The basic circuit for making such measurements is presented in Fig. 17.7. Here, an oscillator generates a voltage $V_1\cos 2\pi ft$ which is applied across a high resistance R_1 in series with the unknown impedance Z. The current flowing in Z produces a voltage $V_1\cos(2\pi ft + \varphi)$ αυ υθε ιξπφυ οζ α θιηθ ινπεδαξγε οπεσαυιοξαμ ανπμιζιεσ (ινπεδαξγε Z_{in}) which forms part of a detector arrangement. The amplified output is chopped in phase with the oscillator voltage

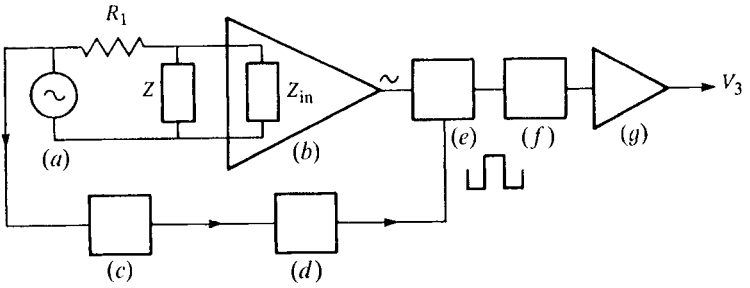


Fig. 17.7 Circuit arrangement for the measurement of the unknown impedance *Z*: (a) oscillator (b) high impedance operational amplifier (c) phase adjuster (d) squarer (e) phase-sensitive demodulator (f) low pass filter (g) DC amplifier; R_1 shunt resistance.

$V_1 \cos 2\pi ft$ by means of a squarer, which operates on a phase-sensitive demodulator that enables the positive in-phase signal to appear at the input of the DC amplifier. The squarer also switches the output of the AC amplifier in opposite phase to the oscillator voltage (push-pull operation) and enables a similar signal to be inverted in the phase-sensitive demodulator to give a full-wave rectified signal

$$V_3 = (A_d V_1 / R_1)|Z| \cos \phi \tag{12}$$

at the output of the DC amplifier, where A_D is the conversion gain of the detector. The purpose of the low pass filter is to exclude unwanted signals from all sources of frequency greater than *f*.

Combining equations (4) and (12) we have

$$V_3 = (A_D V_1 / R_1) R_s$$

It is a simple matter to convert successive readings of V_3 obtained during bacterial growth either to a resistance change $\delta R_s = R_s - R_0$ or to a conductance change $\delta G_s = 1/R_s\ 1/R_0$ by means of an A/D recorder and a computer.

Impedance change δZ or admittance change δY are similarly obtained from readings of V_3 by disconnecting the phase-sensitive demodulator from the circuit and connecting the output of the high impedance operational amplifier directly to the input of the DC amplifier.

17.5.2 Multiplexed cell switching

Automated data logging of multiple cell impedances requires each cell to be connected in turn to a measurement system. Mechanical switching can be employed but this brings with it a risk of variable contact resistance and a limited component lifetime. It is therefore preferable to utilise a system of solid state switches in the arrangement shown in Fig. 17.8, in which measurement is largely independent of switch resistance (typically 100 Ω 'on' and greater than 10 MΩ 'off'). A dual-ganged switch, driven by a suitably timed clock pulse, first connects a stable oscillator to each of the cells in turn via a high value resistor R_1 (typically 100 kΩ) and then feeds the signal voltage produced into a high impedance detector. This, in turn, feeds an analogue-to-digital converter, the output of which is accumulated in a buffer for subsequent processing by a computer. A cell resistance of 1000 Ω produces an output of 10 V DC into the A/D converter. If, for example, this is a 16 bit converter, the resolution of 153 μV corresponds to 0.0152 μS.

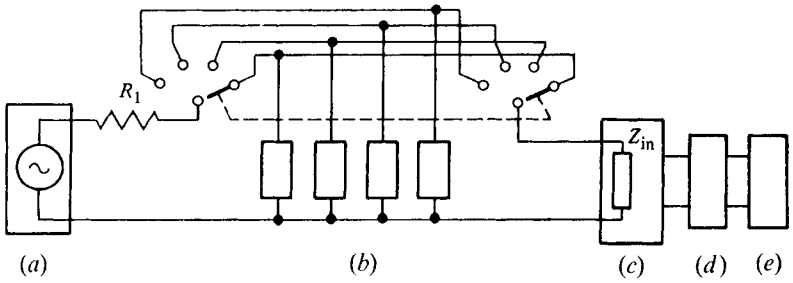


Fig. 17.8 Arrangement for multiplexed cell impedance measurement (solid state switches are chosen): (a) oscillator (b) cells (c) detector (d) A/D converter (e) microcomputer; R_1 shunt resistance.

The zero drift of the electronic circuits is less than 10 μV and the noise is about 10 μV, so that this resolution can be handled with confidence.

The rate of data acquisition is slow in terms of the capability of microcomputers. In fact, the limitation to the speed of switching is imposed by the settling time of measurement, that is the time required after switching for the measuring current to approach its steady value. This is related to the time constant of the cells and circuitry, that is the product of bulk resistance and capacitance. For 100 μV resolution the settling time is of the order of 1 second, so that 240 cells can be scanned every 4 minutes. It is common practice to scan ten times. In most equipment the cell data are initially stored in a buffer and then rapidly read into memory in batches.

17.5.3 Commercial instrumentation.

It is evident that an easily operated method of measuring impedance or its derivatives is required for quality control in the food industry. Electrical detection of the presence of microbes in media or of their growth rate has become well established technology, especially in the food industry where rapid measurement is essential for perishable products. This approach was reinforced at the First International Symposium on Rapid Methods and Automation in Microbiology and Immunology in 1973 (Héden and Illeni 1975a; 1975b), where equipment based on the measurement of the electrical impedance of microbial cultures was described and has since been recognised as having made the most significant commercial advance. Although the method is rapid relative to conventional microbiological analysis it is still slow, taking some hours to obtain a positive result, but the usefulness of the data has been proven and is ahead of that from many other techniques. As will be shown below, there is still considerable scope for much speedier performance. Most users adopt the 'black box' approach: introduce sample into conventional medium, then wait for result. At this level the understanding of the science involved is low, but this does not detract from its usefulness or its accuracy in routine applications, once established.

At the 1973 Symposium, the machines providing automated impedance measurement were the Bactometer (Cady 1975) and the Bactobridge (Ur and Brown 1975). Later Richards *et al.* (1978) described an instrument which became known as the Malthus Growth Analyser (because the plots of conductance change over time were regarded as similar in appearance to the population dynamics curves published by the philosopher Thomas Robert Malthus in 1798). A similar instrument based on the same work is the RABIT. The BacTrac is a four-terminal electrode system.

At present, the Bactometer (Vitek Systems Ltd, Basingstoke, UK), the Malthus (Malthus Instruments Ltd, Bury, UK), the RABIT ((Don Whitley Scientific, Shipley, UK) and the BacTrac (Purkensdorf, Austria) are in production and marketed worldwide. They are highly automated multi-channel instruments containing dedicated microcomputers for processing the measurement of the impedance of up to 500 cells simultaneously. They sit easily on a bench-top or on their own desk-size tables. They offer sophisticated output display of numerical and graphic results and diagnostic information which may be replicated as hard copy by a printer/plotter, or transferred to other computers. The Bactometer and the RABIT have a resolution of 10^6 cfu ml^{-1}, the Malthus 10^5 cfu ml^{-1}. Resolution means the concentration of microbial cells required to cause a significant change in the measured parameter. The true resolution of these instruments is 1 cell, as it will multiply and reach the resolution threshold in due course. The Malthus, RABIT and BacTrac analysers allow the introduction and removal of single samples while in the Bactometer the electrodes are arranged in sets of 8. The importance of close temperature control has been taken into account in the design of the instruments. The Malthus incorporates a water bath incubator (see Fig. 17.9(a)) controlled to ± 2 mK, the volume of water being large enough to minimise any perturbations caused by the addition of cold or hot samples. The RABIT (Fig. 17.9(b)) similarly controls the temperature of individual cells (Fig. 17.10(b)) held in metal blocks. The Bactometer (Fig. 17.9(c)) uses an air incubator with pairs of cells; one of each pair is inoculated and the other serves as a reference. The Bactrac (Fig. 17.9(d)) uses a four terminal electrode arrangement and an aluminium incubator. The instruments each have their own type of electrode design and the cells are not interchangeable. The electrodes used in the Malthus Growth Analyser consist of a small area exposed at the tip of each of two thin platinum strips deposited on a ceramic substrate and insulated (except at the tip) by glass frit. The top of the ceramic substrate is moulded into a plastic cap which is screwed on to standard glass tubes: together they form a conductance cell. The cell can be re-used hundreds of times. They also supply a disposable cell with plastic sheathed tungsten-alloy electrodes. Electrical contact is via the tube cap and thus any size of bottle can be used from 3–100 ml capacity. All the other companies' electrodes make contact through the base. The Bactometer electrodes are formed of two stainless steel strips in the base of each of sixteen wells in a plastic tray; each well holding 2 ml and its electrode constitute a cell, and one electrode of each pair of cells is interconnected so as to constitute a sample cell and a sterile reference cell. They are single use. The RABIT electrodes consist of two stainless steel

(a)

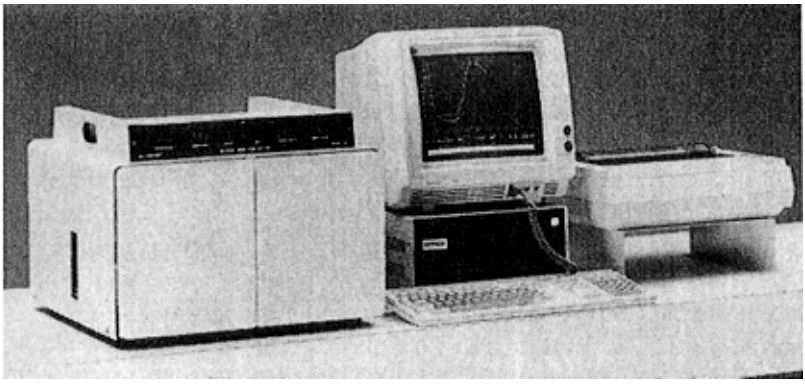


Fig. 17.9 Current commercial instruments (a) Malthus, (b) RABIT, (c) Bactometer and (d) BacTrac.

(b)



(c)



(d)



Fig. 17.9 Contd.

studs protruding through a plastic plug inserted in the base of a cylindrical glass tube. Suitable edge connectors couple the electrodes to the measuring circuitry in each system. They can be re-used. BacTrac has single and multiple use cells but do not disclose their structure or composition. From their literature, there appear to be four electrodes set in the base of a plastic bottle of 20 ml capacity.

The Bactometer displays the change of impedance ratio $Z_r/(Z_r + Z_s)$, where Z_r is the impedance of the reference cell and Z_s is the impedance of the sample or experimental cell, and has the option of displaying the conductance and capacitance normalised similarly (Eden and Eden 1984). The Malthus Analyser measures series resistance at

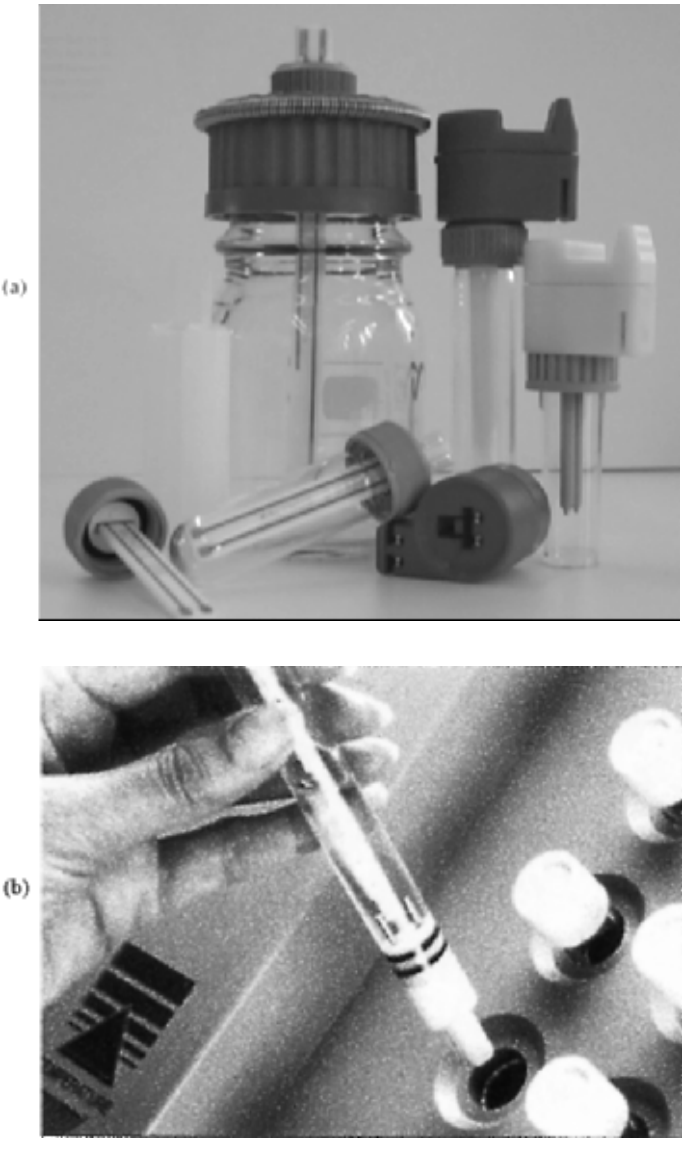
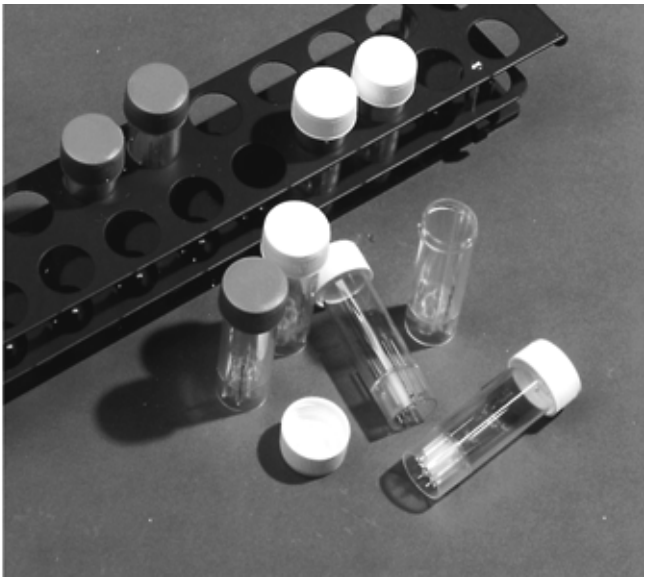


Fig. 17.10 Electrode cells for each instrument (a) Malthus, (b) RABIT, (c) Bactometer and (d) BacTrac.

(c)



(d)

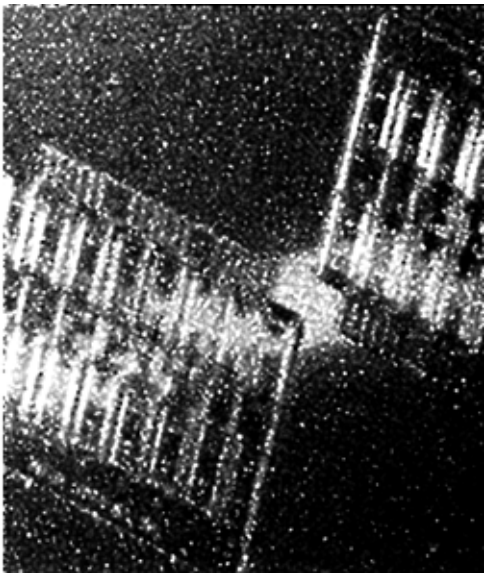


Fig. 17.10 Contd.

10 Hz and displays conductance change as the contribution from capacitance is minimal at this frequency. The RABIT measures at 2 Hz and has more capacitance in the impedance values. The Bactometer can be used in conductance, capacitance or impedance modes. The BacTrac is called an impedance splitting device by its manufacturers; it displays the relative medium impedance (M value) or the relative electrode impedance (E value). As changes in the electrode impedance are significant, the electrodes are probably made of base rather than noble metals. There is also a 21 sample version that measures only the solution impedance.

The importance of measuring series conductance in high resolution growth determination has been emphasised earlier, and it has been pointed out that this must

be coupled with the use of platinum electrodes and close temperature control. Only the Malthus offers this combination, though it must be emphasised that this is of advantage only when very early detection of growth is necessary. The Malthus, RABIT and Bactrac can be used in the indirect conductance mode, that is, the electrodes are placed in an alkaline solution separate from the growth medium from which CO_2 diffuses.

17.6 The evaluation of conductance data

17.6.1 Bacterial growth in batch culture

Consider a population of initial density n_0 which commences to grow in a medium of mass concentration c at lag time t_L after inoculation. Each bacterial cell metabolises an equal mass ΔS of the substrate during growth, and growth eventually ceases owing to the accumulation of a limiting concentration of toxic metabolites when the total number of cells present is n_s. The number of cells that have grown in this period is

$$n_s - n_0 = (S_0 - S_s)/\Delta S$$

where S_0 and S_s are the initial and stationary values respectively.

The initial specific growth rate is $(\dot{n}/n)_0$, where $\dot{n} = \mathrm{d}n/\mathrm{d}t$ is constant for a given strain growing in a given medium at a given temperature (Jason 1983). It has the dimension of $(\text{time})^{-1}$, so that we write

$$t'_g = 1(\dot{n}/n)_0 \tag{13}$$

where t'_g is characteristic of the reproduction and growth of a bacterium in the absence of any self-generated inhibitory substance, n is the population density at time t and $\dot{n}$ signifies the growth rate $\mathrm{d}n/\mathrm{d}t$.

The specific growth rate declines from its initial value to zero when $\dot{n} = n_s$. Furthermore, $(n_s - n_0) \propto (S_0 - S_s)$, so we may assume that

$$\dot{n}/n = (\dot{n}/n)_0[1 - (n - n_0)/(n_s - n_0)]$$

Thus

$$(\dot{n}/n)/(\dot{n}/n)_0 = (n_s - n)/(n_s - n_0) \tag{14}$$

Combining equations (13) and (14) we have

$$\dot{n} = (n/t'_g)[(n_s - n)/(n_s - n_0)] \tag{15}$$

On integration we obtain the relationship between population growth and time t after inoculation. Integration is over the limits $[n_0 \rightarrow n]$ and $[t_L \rightarrow t]$ and we approximate $n_s \pm n_0 \simeq n_s$ and write in $\dot{n} = \mathrm{d}n/\mathrm{d}t$.

$$n = n_0 n_s/\{n_0 + n_s \exp[(t_L - t)/t'_g]\} \tag{16}$$

The sigmoidal curve associated with monophasic growth (shown in Fig. 17.3) is closely followed by this equation, where the constant of proportionality (equation (9)) is 1 pS(cfu ml^{-1})$^{-1}$. Equation (16) thus scaled is represented by the broken line in the figure.

Equation (16) reduces to the familiar exponential form for the early stages of growth.

$$n = n_0 \exp[(t - t_L)/t'_g] \tag{17}$$

when the approximation $n \ll n_s$ is used in integrating equation (15). The generation time t_g is defined as the time during which the population doubles during the initial stages of growth, that is when the population is well below the saturation level ($n \ll n_s$). It therefore follows from equation (17) that

$$t_g = t'_g ln2 \tag{18}$$

17.6.2 Determination of inocula

Growth in bacterial density follows the relationship of equation (16), but often suddenly changes a few generations after the peak growth rate has been reached. The culture then continues to grow at a slower rate which is also described by equation (16) but with a larger value of t_g and with other parameters changed. There can be several such growth phases, but for present purposes only exponential growth in the first phase is considered.

Rearranging equation (17)

$$t - t_L = t'_g \ln(n/n_0)$$

Substituting for t'_g from equation (18) and writing $\ln(n/n_0) = 2.303\lg(n/n_0)$

$$\lg\, n_0 = \lg\, n - 0.301(t - t_L)/t_g \tag{19}$$

If the detection time t_D is the time at which a batch culture grows to an arbitrary fixed population density n_D, then the relationship between $\lg\, n_0$ and t_D is linear:

$$\lg\, n_0 = \lg\, n_D - 0.301(t_D - t_L)/t_g \tag{20}$$

Therefore, this is the means of determining an inoculum number of a given organism from a previously obtained calibration simply by finding the time at which the conductance change ΔG_D, corresponding to n_D, occurs and by reading off the value of n_0. It is the basis of determining bacterial counts from the detection time, the time at which a significant change of conductance/impedance relative to a control occurs. The control can be any uninoculated cell or, as with the Bactometer, a paired cell for each assay.

The generation time of the organism is

$$t_g = -0.301/m$$

where m is the slope of the line.

It is customary to select a value of ΔG_D corresponding to about 10^6 cfu ml^{-1}, but its magnitude is unimportant as long as t_D can be determined accurately from the growth curve to within ±0.1 h.

The calibration for pure cultures is best obtained from the determination of a series of values of t_D corresponding to values of n_0 obtained from successive dilution of a parent inoculum of high concentration. In this way, the scatter inherent in the determination of successive individual values of n_0 by plate counts is avoided. The statistical error implicit in obtaining a plate count of N colonies grown from a sample volume of a true inoculum n_0 is $\pm N^{1/2}$ or $\pm 100N^{1/2}/N$ per cent, an error that remains unchanged whatever the dilution of the parent inoculum. From a large value of N (say > 100), the error in determining the daughter inocula is insignificant in relation to the parent value, being attributable only to the small error of dilution. A calibration curve derived in this way has no detectable scatter (Fig. 17.11). In most situations, however, mixed cultures are present in foods and

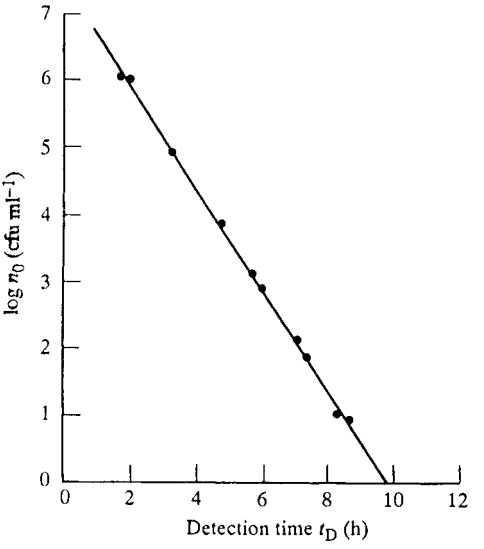


Fig. 17.11 Relationship between inoculum concentration n_0 and detection time t_D for *E. coli* grown in Oxoid nutrient broth at 37°C.

the data are more scattered (Fig. 17.12). This has the effect of introducing an uncertainty in estimating the inoculum concentration from the value of a specific detection time.

The degree of uncertainty is calculated from the standard error of estimate (SE_{lnn0}) of lg n_0 on t_D. Its value is given approximately by the expression

$$SE_{lnn0} = \sigma_{lgn0}(1 - r^2)^{1/2}$$

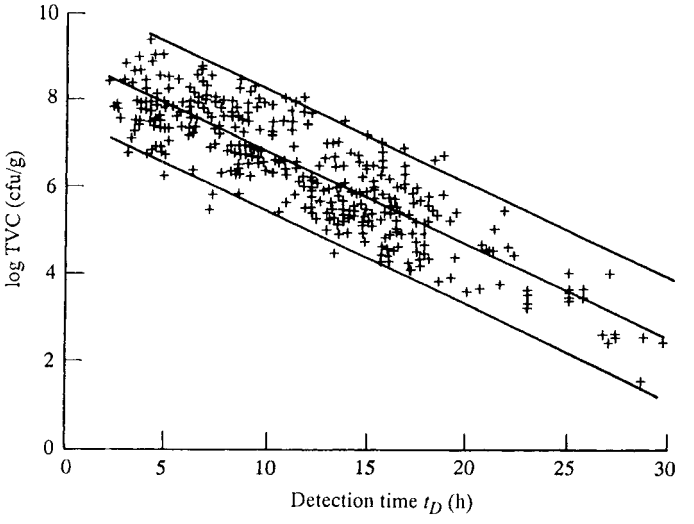


Fig. 17.12 Relationship between total viable count (TVC) of various fish products (cod, haddock, plaice, trout and scampi) and detection times (t_D assayed in brain heart infusion at 20°C: the 95 per cent confidence limits are indicated (from Gibson and Ogden 1987, by courtesy of the authors and publisher).

where $\sigma_{\lg n0}$ is the standard deviation of all lg n_0 values and r is the correlation coefficient. Thus 95 per cent of all values lie within plus or minus two standard errors of estimate of the values given by the regression equation, and practically all values lie within plus or minus three standard errors of estimate.

17.6.3 Choice of growth media

In many assays for growth of microbes in undefined media that contains protein extracts, beef extract, yeast hydrolysates, sugars, etc., the conductance change cannot be ascribed to any particular ions, compounds or reactions. In some cases the main source is known and has been confirmed by chemical analysis of fresh and spent media, or by obtaining a quantitative response to altering the concentration of the suspected chemical. For example, Easter *et al.* (1982) showed that the reduction of trimethylamine oxide, an uncharged molecule, to trimethylamine, a very basic molecule, resulted in a quantitative conductance change (Fig. 17.13); the compound has been used in assays for fish quality and for assays for salmonellas for food safety. Owens and Wacher-Viveros (1986) suggested that organic buffers could be employed, and this approach has been particularly useful in media formulations for yeast assays. Connolly *et al.* (1988) used ammonium tartrate and Adams *et al.* (1989) included succinate buffer to good effect. The ionisation of inorganic ions may change during the course of the assays owing to changing pH and may affect the conductance (Owens *et al.* 1985; Gibson 1990).

Most studies on the source of the conductance change have been done with pure cultures in defined media. In the analysis of foods and beverages, there is a significant nutritional contribution to the medium from the sample being tested, and this can influence the composition of the medium and the metabolism of the organisms present. Indeed, to obtain data on the stability or spoilage of products, Gibson (1990) has argued that the medium should contain the constituents of or reflect the composition of the material under test.

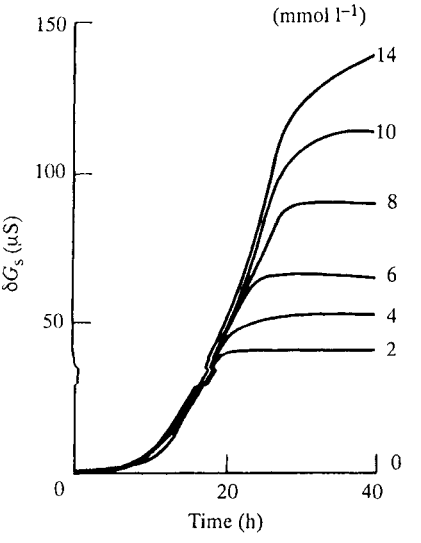


Fig. 17.13 The effect of trimethylamine oxide (TMAO) concentration on conductance change δG_s: *Shewanella putrefaciens* NCMB 400 was grown in broth containing 0–14 mmol l^{-1} TMAO (from Easter *et al.* 1982, by courtesy of the authors and publisher).

For special applications, a measurement of evolved carbon dioxide by a variant of the conductance method has been suggested. During growth, all micro-organisms produce CO_2. Numerous ways to measure this have been proposed (Dixon and Kell 1989) including some conductimetric methods (Hill and Evans 1986; Bruckenstein and Symanski 1986). Linear quantitative responses were found over a range of concentrations of up to 100 mM. Owens *et al.* (1989) modified Malthus and RABIT electrodes for CO_2 measurement. They placed the electrodes in a solution of KOH or NaOH which was kept separate from the growth media. Carbon dioxide produced by growth was trapped in the alkali, causing a change in conductance. The response with *E. coli* was slower than by direct conductance assay, but Owens *et al.* propose its use for assays of microbes which give a poor response by direct assay. There could of course be interference by CO_2 produced by tissue respiration. As chemical solutions of alkalis are conductive with no capacitance, simpler measurement systems than impedance can be used.

17.6.4 Correlation with conventional microbiological data

Conductance data are used in place of conventional microbiological data for three purposes, to assess the quality of products, their safety and also to predict their shelf life. Hitherto, quality has usually been derived from the standard plate count (TVC or cfu ml^{-1}) and, while it is one of the most used assays in the food industry, it is of limited value in quality control or assurance, with results available days after sampling. Sharpe (1980) has said that 'no other analytical microbiological procedure in the scientific world can correlate with it' and this is still true. Thus it is not possible for any novel technique, regardless of its scientific merits, to give data fully comparable with the conventional assay; indeed, it is not desirable that it does so, for the plate count has many drawbacks. It is a measure of the ability of microbes to form colonies or biomass on the test medium.

Theoretically each microbe gives rise to a colony, but it is well known that colonies can arise from clumps of perhaps hundreds of cells. In a liquid assay, each viable cell in the inoculum has an individual contribution and, even if it is not replicating, its metabolism end products may contribute to the conductance change. Thus the magnitude of correlation coefficients between conventional and conductance or impedance data is not necessarily a valid criterion for the acceptability of the latter. Calibration curves relating plate counts and results from other conventional methods to, say, detection times have been published by many authors. Usually they cover a wide range of microbiological quality, far greater than is found for particular products in commerce. Samples are modified to give a range of qualities, for example by holding for longer than usual at higher than normal temperatures. While such procedures were acceptable in the early days of automated microbiology, they are not needed now. Accept/reject decisions can be based on the actual detection time without recourse to the calibration curve and conversion to microbial numbers, by comparison with a target detection time varying with product and desired quality. Of high importance is the reproducibility of the assay, allowing for the uneven distribution of microbes in many materials and products.

In conventional assays involving homogenisation or extraction, dilution and plating, the amount of starting material actually assayed is in the milligram range. In electrical assays, only a primary dilution is needed (to eliminate antimicrobial effects of the constituents of the sample), and in an assay of 10 ml volume, gram quantities can be taken. It would be reasonable to assume that it is better to analyse larger rather than smaller quantities of heterogeneous materials. Malthus Instruments produce a cell with a 100 ml capacity so an assay on 10 g of food is quite feasible. As most conventional assays

test only 20 mg of the food (Gibson, 1990) there are statistical implications of taking such a larger sample, as for example as a composite from many subsamples, especially for safety.

Table 17.3 lists some products and product categories that are known to be analysed by electrical methods in commerce. These probably represent the main use of the technique. There are many other protocols described in conference abstracts and in manufacturers' literature. The safety of foods relates to the absence of specific bacteria or groups of organisms. Assays have been developed for the organisms listed in Table 17.4. The electrical assays have been particularly useful in this respect. Many conventional assays are slow, laborious and even contentious; for example, the effects of different selective agents for salmonellas are disputed. Conductance assays have been accepted for use in industry to give rapid clearance (or otherwise) of raw materials and products. Indeed, it has been said that the only economic justification in the food field for such expensive equipment is product safety. There are many competing methods, especially in single use kit form as well as more sophisticated methods such as the DNA analysis, but while they may offer advantages in specificity, they are no faster than the electrical assays as they normally require prior growth of a culture that takes most of the overall assay time (Gibson, 1998).

As an example of the use of conductance measurements in the regulatory field, the authorities in France responsible for shellfish hygiene use Malthus instruments located in laboratories around their coastline for ensuring that harvested bivalve shellfish meet EU regulations. They carried out extensive trials (Dupont *et al*., 1994) and devised their monitoring system (Dupont *et al*., 1996). The reference method is the Most Probable Number (MPN), rather than a straightforward count and the objective of the measurements is to grade the shellfish rather than to have a continuous spectrum of qualities. The total assay time is 9 hours, specimens yielding longer detection times are automatically allocated to the top grade. The MPN method takes three days. Thus, any

Table 17.3 Some applications of conductance and impedance methods for food quality

Dairy products	
Raw milk	O'Connor (1979); Gnan and Luedecke (1982); Firstenberg-Eden and Tricario (1983)
Pasteurised milk	Visser and de Groote (1984); Kamei *et al*. (1988)
Dried milk powder	McMurdo and Whyard (1984); Prentice and Neaves (1987)
Fermented milk	Okigbo and Richardson (1985)
Cream	Griffiths and Phillips (1984)
Cheese	Waes and Bossuyt (1984)
Fish	Gibson *et al*. (1984); Ogden (1986); Jorgensen *et al*. (1988); Delgaard *et al*. (1996a)
Brewing	Evans (1985)
Wine	Henschke and Thomas (1988)
Vitamins	Einarrson and Snygg (1986)
Meat	Firstenberg-Eden (1983); Bulte and Reuter (1984)
Vegetables	Hardy *et al*. (1977)
Confectionery	Pugh *et al*. (1988)
Fruit juices	Jarvis (1985)
Shelf life	Gibson (1985); Gibson and Ogden (1987); Jorgensen *et al*. (1988); Lindberg and Borch (1994); Delgaard *et al*. (1996a)
Quality	Bollinger *et al*. (1994); Schulenburg and Bergann (2000)
Antimicrobials	Tassou *et al*. (1995)

Table 17.4 Conductance and impedance methods for microorganisms important in food safety

Salmonella spp.	Easter and Gibson (1985)
	Gibson (1987)
	Arnott *et al*. (1988)
	Ogden (1988)
	Pugh *et al*. (1988)
	Bullock and Frodshaw (1989)
	Pless *et al*. (1995)
	Quinn *et al*. (1995)
E. coli and coliforms	Silverman and Munoz (1979)
	Martin and Selby (1980)
	Gibson *et al*. (1984)
	Dupont *et al*. (1994)
	Colquhoun *et al*. (1995)
	Dupont *et al*. (1996)
	Timms *et al*. (1996)
	Gibson (1997)
	Edmiston and Russell (1999, 2000)
Enterobacteriaceae	Petitt (1983)
	Wood and Williams (2000)
Clostridium spp.	Dromigny *et al*. (1997)
Clos. botulinum	Gibson, A. M. (1987)
Histamine producers	Klausen and Huss (1987)
Faecal streptococci	Neaves *et al*. (1988)
Listeria spp.	Phillips and Griffiths (1989)
	Bolton and Gibson (1994)
Mycotoxigenic moulds	Adak *et al*. (1987)
Yeasts	Connolly *et al*. (1988)
	Schaertel *et al*. (1987)
	Deak and Beuchat (1994)
Bacteriophage	Waes and Bossuyt (1984)

products falling into the reject grade can be removed from commerce before it reaches the market. Much of this work was confirmed in a multinational EC-funded project (Ogden *et al*., 1998) where there was high correlation between the grades given by conventional methods and the Malthus method.

In an industrial context, Wood and Williams (2000) have described how they use conductance assays to reduce the level of Enterobacteriaceae in animal feed production. They already use Malthus instruments for the control of *Salmonella*. Normal processing reduces the numbers of these organisms but where recontamination from any source occurs, the rapid assays allow remedial action to be taken.

It has rarely been possible to take conventional media and use them in electrical assays as they have been formulated for different objectives. Media components for plate counts are selected for clarity and low colour so that colonies stand out from the background. An example is the assay for *Salmonella* spp. in foods. Easter and Gibson (1985) tested most of the commonly used media for salmonellas (which incidentally do not give equivalent results). They found that many were outside the measuring range of the conductance instruments, due to their high ionic strength, and others failed to give a distinct conductance change. In their work on fish spoilage, Easter *et al*. (1982) had observed that the reduction of trimethylamine oxide (TMAO) gives a large conductance change, and it was evident from the literature that the ability to reduce TMAO was common to salmonellas. They added TMAO to the most promising media, selenite-cystine broths,

and produced, with other modifications, a medium in which most salmonellas are detected, along with some *Citrobacter freundii*.

Alternative media have been formulated that are based on the ability of salmonellas to decarboxylate lysine (Arnott *et al.* 1988; Ogden 1988). All these media have been tested extensively under industrial conditions and it can be concluded that their performance is very good – better than many media used in conventional assays, and only limited in reliability by the statistics of sampling. The original Easter-Gibson formulation has been tested in a ring trial run on behalf of the International Dairy Federation on 2500 samples (Prentice *et al.* 1990). Smith *et al.* (1989) have tested it, along with Ogden's formulation, on animal feeds and their method has statutory status in the UK for testing of animal feedstuffs. The Easter-Gibson medium and Ogden's medium have been granted AOAC approval for testing for *Salmonella* spp in all foods (Gibson *et al.*, 1992) and are a British Standard method BS 4285 subsection 3.9.2 (1992). With the upsurge in the requirements for testing for salmonella (and other pathogens), laboratories can cope with the workload only by using automated methods such as described here.

17.7 Future possibilities

A criticism frequently levelled at the application of impedance techniques in microbial assay is the long period before detection, even under the most favourable conditions. The detection time, of course, depends on the inoculum concentration and on the arbitrary level of conductance change for detection (Fig. 17.13). In this example, a detection level is registered two hours after inoculation with 10^6 cfu ml^{-1} and eight hours after inoculation with 10^{-1} cfu ml^{-1}. It is clearly desirable to improve this performance for more speedy results.

In the absence of baseline drift, growth is detected when the change in conductance exceeds ΔG_s (as defined in equation (10)). This occurs after the elapse of γ_D generations of growth of any given organism, and it may be shown that

$$\gamma_D = 3.32(\log\Delta n - \log n_0) \tag{21}$$

where Δn corresponds to ΔG_S and n_0 is the inoculum concentration. Reference to equation (10) shows that when $R_s = R_f$, a 16 bit A/D voltage converter resolves conductance to within 0.015 μS, corresponding to a numerical resolution of 1.5×10^4 cfu ml^{-1}. Electronic drift and noise are equivalent to less than one-tenth of this resolution, so that without resorting to elaborate measures it is possible to anticipate a resolution of about 10^3 cfu ml^{-1}. Thus, taking a value of $\Delta n = 10^3$ cfu ml^{-1} and assuming that a detection level of ten times this resolution denotes the detection time, we can predict the number of generations that must elapse before growth is detected for a range of inocula (Table 17.5). Therefore it seems reasonable to expect almost immediate detection of the growth after the lag phase when the inoculum is $> 10^4$ cfu ml^{-1}. If the inoculum is 10^3 cfu ml^{-1} and the generation time is 0.3 h, then detection would occur one hour later. Thus, by using a cell of resistance R_s close to the maximum value R_{fs} and a 20 bit A/D converter, a very early detection (following the lag phase) would be possible for determining quite low bacterial loads. Such a possibility represents only a small design effort on the part of manufacturers.

Cell design, too, is likely to be improved. Electrodes, for example, could be screened from deposition of detritus from the food sample which will lead to a more stable

Table 17.5 Threshold number of generations γ_D obtainable on 16 bit A/D converter for various n_0 ($R_s/R_{fs} = 0.9$)

n_0	10^4	10^3	10^2	10^1	10^0	10^{-1}
γ_D(no.)	0	3.3	6.6	10.0	13.2	16.6

baseline. Thus earlier detection will be possible with samples that are at present difficult to monitor. It is also advantageous to develop media that give rise to the shortest possible lag phases and generation times, since detection times are correspondingly reduced. They do not need to be as productive as in conventional assays as the growth is detected electrically before it is visible.

In many countries the expected shelf life, during which the food should be of good quality, has to be marked on packages. Generally a considerable error margin is allowed for in the absence of data, and this can lead to losses of perfectly edible food. For foods in which microbes are the main cause of deterioration or spoilage, rapid assays can yield data relating to the predicted shelf life of the products. Gibson (1985), Gibson and Ogden (1987) and Jorgensen *et al.* (1988) have used conductance detection times for this purpose for fish. They based their method on the time-temperature relationships of Ratkowsky *et al.* (1983) for microbial spoilage at chill temperatures. The method is promising and, with some refinement, could lead to an electrical assay indicating not only the quality at the time of testing but also the expected shelf life at any selected storage temperature. It should be supplemented with careful monitoring of the storage temperature or the use of time-temperature integration labels. As details of the sample, its origin and test results are already stored in the computer, the method has great potential for the traceability of foods, a legal requirement in the near future.

The cultures grown in conductance assays have been used as the source of bacteria for some of the molecular assays described in Section 17.2. They represent screened positives, obtained at low cost. Rapid methods utilising expensive reagents or considerable 'hands on' time can then be justified and can give results in a very short time as the pre-incubation step has been done. Proponents of many tests that are said to take only a few hours forget to include this step and Gibson (1998) has pointed out that the total time of, for example, some PCR tests is very similar to that for conductance assays.

The possibility of identifying bacteria from the shape of the conductance curve, or from the plot of the rate of change of conductance with time since inoculation has been explored by Jason and Jason (1998). They have convincing evidence that with suitable media, species and sub-species can be identified by curve analysis so that a contaminant in a food can be simultaneously detected and identified.

To conclude, conductance/impedance methods have become established as accepted conventional methods in food microbiology and have many international approvals. They involve the minimum of technical skill, effort and intervention, being in effect 'hands-off' or 'walk away', and give results in a quarter of the time taken by traditional tests. As larger quantities of food can be put into the actual test compared with traditional tests, they should give more sound results. Some of the emerging methods can be linked to conductance methods, for example, immuno-magnetic separation to increase the microbial content of the inoculum, and colour detection by the specific metabolism of chromogenic substrates. The instruments have become very reliable over the years and advances in computing and software have improved the presentation of results. Further advances from specific growth medium engineering can be expected. Electrical assays

have moved on from simply replacing conventional assays to being the prime data generators for emerging techniques such as predictive microbiology and this trend will continue.

17.8 References

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